Revisiting SLIC: Fast Superpixel Segmentation of Marine SAR Images Using Density Features

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***Abstract*— The simple linear iterative clustering (SLIC) has been shown as an efficient and widely used superpixel-based algorithm for segmenting marine synthetic aperture radar (SAR) images. However, SLIC does not consider the fact that the density of ship target pixels is significantly lower than that of sea clutter pixels, leading to a waste of computational cost and memory resources on lots of pure clutter areas and to the degradation of the compactness of superpixels. To address the aforementioned issues, we develop a new density-based SLIC (DSLIC) method for the superpixel-based segmentation of marine SAR images. In the initialization stage of our DSLIC, all the subimages in a large marine SAR image are rapidly prescreened via a new density-driven classifier, where most of the subimages only occupied by clutter pixels with comparatively high density are discarded and do not need to be segmented in the subsequent local clustering stage. The retained subimages contain both the clutter and potential target areas. This prescreening operation results in higher computation efficiency and memory savings. In the local clustering stage of DSLIC, besides the intensity proximity and the spatiality proximity (used in SLIC), the sparsity proximity (measured by density distances) is considered to reduce the coexistence of sparse target pixels with low density and nonsparse clutter pixels with high density within superpixels. Our theoretical and experimental results show that the proposed DSLIC method is faster and requires less memory than SLIC and other state-of-the-art superpixel-based segmentation methods for marine SAR images with similar or better segmentation accuracy.**

***Index Terms*— Density, ship targets, superpixel-based segmentation, synthetic aperture radar (SAR).**

I. Introduction

SHIP detection and monitoring in marine areas play an important role in many civilian and military applications, including sustainable fishery, traffic management, and military reconnaissance [1]–[3]. Synthetic aperture radar (SAR) is an active microwave sensing technology that can generate high-resolution images with large swath widths in all the weather and illumination conditions [3]–[6] and has been extensively applied for ship detection and monitoring [10], [12], [39].

A. Superpixel-Based Segmentation of Marine SAR Images

As the increase in resolution of marine SAR images, a ship target therein often occupies hundreds or thousands of pixels with similar attributes in the local region and appears as bright pixels mainly due to the double reflection of radar pulses [33]. Superpixels contain a number of locally aggregated pixels and have been widely used as basic processing units in high-resolution SAR images [7]–[14], [23]–[26], [50]. Compared with pixel-level processing, the use of superpixels exhibits the following advantages for marine SAR image processing.

1) Superpixels are helpful to accelerate the implementation of ship target detection because of the significantly reduced number of processing units [7].
2) It is generally more convenient to extract region-based features of ship targets via superpixel-based boundary analysis [10], [16], [26].
3) Superpixels can help suppress speckle noise in SAR images [46].

Therefore, superpixel-based segmentation is often used as the preprocessing process of the task of marine target detection in high-resolution SAR images. In the last decade, some research works have been devoted to superpixel-based segmentation of SAR images based on graph theory [15]–[17], mean-shift [18], [19], and local clustering strategies [20]–[24].

Graph-based superpixel segmentation methods treat each pixel in an image as a node in a graph, where the weight of the edge between two nodes represents the similarity degree of the corresponding two pixels. Superpixels are generated by minimizing a cost function derived from the information entropy [15] or normalized cuts [16] defined over the global graph. In [17], multiscale superpixels are generated based on a minimum spanning tree (MST), where the size of superpixels in homogeneous background areas is much larger than that in areas with dense content. Graph-based segmentation approaches maintain the edge information well in SAR images [15]–[17] but often lead to high computational complexity due to the optimization of graph parameters.

Mean-shift-based segmentation methods mainly rely on the identification of the local maxima of probability density

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function of pixels and have been successfully applied in speckle reduction and texture preservation techniques for SAR images [18], [19]. However, they also suffer from a slow segmentation speed due to the need to conduct a global search for the optimal superpixel centers.

Superpixel-based segmentation methods using local clustering strategies have received much attention due to their good boundary adherence and high efficiency in terms of computation and memory savings [7], [20]–[24], [51]–[54]. The simple linear iterative clustering (SLIC) is a well-known local clustering-based superpixel segmentation method [7]. SLIC [7] basically includes two stages: 1) initialization of superpixel centers and 2) local k-means clustering. In the initialization stage, superpixel centers are densely allocated by sampling pixels with regular grids throughout the SAR image. Then, the local k-means clustering is performed to update superpixel centers and assign pixel labels based on the intensity and spatiality proximity of pixels. Recently, several superpixel segmentation methods based on SLIC have been proposed to enhance the segmentation performance of SAR images [20]–[24], [51]–[54]. In [20] and [51], improved initialization approaches for SLIC are proposed to obtain robust cluster centers by exploiting the edge information in images. Instead of the pixel-based intensity proximity used in SLIC, the patch-based intensity proximity (with Gaussian filtering) is considered in [21] to maintain the edges of SAR images in the presence of strong speckle noise. The Euclidean spatiality proximity in SLIC is replaced by the Mahalanobis distance in [22] to better model complex texture structures in SAR images. In [23], the multiorder features introduced by Fisher vectors are embedded into the local k-means clustering stage in SLIC to improve the segmentation performance in SAR images with low target-to-clutter contrast. In [52], a likelihood function established by generalized gamma distributions of pixels is used to reduce segmentation errors of SLIC. In [53], polarimetric homogeneity measurement is used to adaptively adjust the weights of features utilized by SLIC. In [54], the global structure features in SAR images are considered to improve the segmentation performance of SLIC.

B. Challenges

The aforementioned SLIC and its extensions still face two challenges when conducting superpixel-based segmentation of marine SAR images.

1) Note that, in the scenarios of onboard implementation of SAR ship monitoring in satellites, the available computation power and memory resources are stringently limited due to the constrained weight and volume of satellites and the harsh space environment [57], [58]. For SLIC, one of the challenges is the waste of computation and memory resources on large sea clutter regions caused by the dense initialization of superpixel centers. After the dense initialization, SLIC-based methods have to elaborately assign pixel labels throughout the whole image to generate superpixels. However, as shown in Fig. 1, ship targets of interest in large marine SAR images (with open sea areas and swath widths [48] of hundreds of kilometers) occupy only a small number of areas [56], and it is unnecessary to finely segment most of the sea clutter areas therein. Accordingly, SLIC and its extensions suffer from low computation efficiency and high memory requirements when applied to segment marine SAR images (despite their superiority in terms of computation and memory savings compared with commonly used graph or mean-shift-based methods [7]).
2) Another problem is the degradation of compactness of superpixels, caused by the lack of sparsity proximity in the local k-means clustering stage. As we can see in Fig. 1(b) and (c), ship target pixels exhibit sparsity (from a global perspective), while sea clutter pixels occupy most of the regions. The lack of sparsity proximity of pixels in the local k-means clustering stage of the popular SLIC-based methods may lead to incompact superpixels with sparse target pixels and nonsparse clutter pixels coexisting therein.

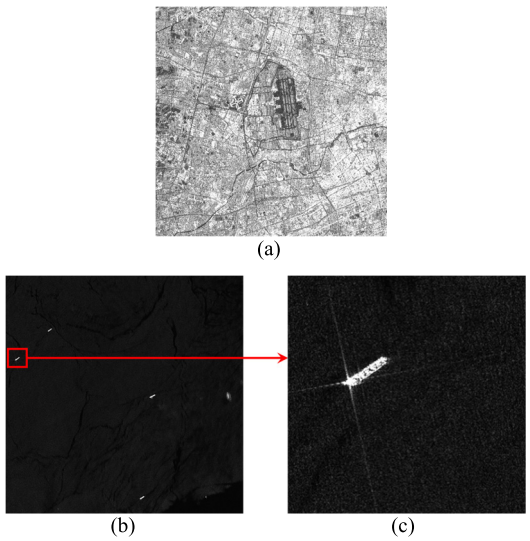


Fig. 1. Comparison of (a) SAR image collected over an urban area (500 × 500 pixels) [32], (b) marine SAR image (3000 × 3000 pixels) with 3 m × 3 m resolution [42], and (c) subimage of (b) (300 × 300 pixels). Unlike urban SAR images, target pixels of interest in marine SAR images show sparsity in terms of spatiality.

The density value of a pixel reflects the number of pixels that are similar to it in an image [27], i.e., density implies the sparsity degree of a given pixel. Density features have been successfully applied in the context of face recognition [28], hyperspectral image processing [29], [30], and biological data processing [31]. To the best of our knowledge, density features have never been exploited to enhance superpixel-based segmentation performance in marine SAR images.

C. Main Contributions

In this article, we propose a new density-based SLIC (DSLIC) method to address the aforementioned challenges that SLIC-based algorithms face when segmenting marine

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SAR images generated by spaceborne sensors. In our newly developed DSLIC, the sparsity degree of pixels is measured by using their density values, where sparse target pixels have significantly lower densities than dense sea clutter pixels. In summary, the main innovative contributions of DSLIC are twofold.

1) First and foremost, we introduce a new initialization method based on density features to reduce the computational cost and memory requirements devoted to noninformative background clutter areas. In the initialization stage of DSLIC, the large marine SAR image is first segmented into a few subimages. Then, a density-driven classifier trained by a weighted sparse least-mean-square (WSLMS) strategy is developed to prescreen each subimage. Only a small number of candidate subimages potentially containing ship target pixels (with significantly lower densities) are delivered to the subsequent local clustering stage. Most of the subimages with pure sea clutter will not be segmented, and the computation efficiency and memory savings are significantly improved. This is helpful to achieve the subsequent real-time SAR ship detection and classification in onboard satellites. Note that the retained candidate subimages still contain clutter areas that can be used to extract the statistical characteristics of the sea clutter.
2) In addition, an improved local k-means clustering method is developed, where the sparsity proximity (represented by the density distance between two pixels) is newly considered in addition to the commonly used intensity proximity and spatiality proximity in SLIC. This is helpful to enhance the compactness of superpixels by clustering sparse target pixels and nonsparse clutter pixels, respectively.

The theoretical analysis of computation and memory savings of DSLIC is provided to show its superiority in comparison with SLIC [7]. Our experiments based on large marine SAR images (from Gaofen-3 and Sentinel-1 satellites with open sea areas) demonstrate that the proposed DSLIC method provides higher computation efficiency than other commonly used algorithms with comparable or better segmentation performance. Furthermore, our DSLIC leads to dramatically faster implementation of superpixel-based ship detection without noticeable detection performance loss.

D. Article Organization

The remaining of this article is organized as follows. Section II briefly introduces the SLIC algorithm [7]. In Section III, we describe our new DSLIC algorithm for superpixel-based segmentation of marine SAR images. Section IV theoretically analyzes the computational complexity and memory requirements of our DSLIC. Section V provides in-depth comparisons between the proposed DSLIC approach and other existing methods using detailed experimental assessment. Section VI concludes this article with some remarks and hints at plausible future research.

II. Background: SLIC Algorithm

Let us assume that the number of pixels in a marine SAR image is given by N, and the expected number of pixels in a superpixel is given by S^2. S denotes the superpixel size. Let (x_i, y_i) and δ_i denote the spatial coordinate and the intensity value of the ith pixel, respectively, for $i = 1, 2, \ldots, N$. In SLIC, the proximity $D(i, j)$ between pixel i and pixel j is defined as

$$D(i,j) = \sqrt{\left(\frac{d_{i,j}^{(\text{spat})}}{\tau_1}\right)^2 + \left(\frac{d_{i,j}^{(\text{inte})}}{\tau_2}\right)^2} \tag{1}$$

where $d_{i,j}^{(\text{spat})} = ((x_i - x_j)^2 + (y_i - y_j)^2)^{1/2}$ and $d_{i,j}^{(\text{inte})} = |\delta_i - \delta_j|$ represent the spatiality proximity and intensity proximity, respectively, for $i, j = 1, 2, \ldots, N$, and $\tau_1, \tau_2 > 0$ are two normalized factors. In SLIC, the superpixel centers are first densely initialized based on regular grids with an interval S. Then, the local k-means clustering operation based on the proximity in (1) is performed to update superpixels in SAR images. Finally, we conduct postprocessing to enhance the local connectivity of superpixels.

III. Proposed DSLIC Algorithm

In this section, we introduce the initialization and local clustering stages of our new DSLIC method that exploits density features of marine SAR images. An overview of the proposed DSLIC method and a comparison between DSLIC and SLIC [7] are provided in Fig. 2.

A. Stage 1: Initialization Based on Density-Driven Classifier

The swath widths of marine SAR images collected by spaceborne satellites can reach hundreds of kilometers [47], [48], while only a few subimages may contain ship targets. In the task of ship detection and monitoring, we are not interested in most of the pure sea clutter subimages, so they do not need to be finely segmented. In this section, we introduce the proposed initialization method for our DSLIC algorithm that prescreens all the subimages in a large marine SAR image based on a new density-driven classifier. This prescreening operation discards most of the subimages with pure clutter and optimizes the computation efficiency and memory requirements of the subsequent local clustering stage. This initialization step performs the following operations.

1) Acquisition of Subimages: We first uniformly segment a large marine SAR image with N pixels into several $M_1 \times M_2$ subimages [as shown in Fig. 1(c)], where $M_1 \times M_2 \gg S^2$, target pixels still show sparsity in terms of spatiality, and clutter pixels are still dense in each subimage. A small overlap between adjacent subimages can be set to avoid the scenario that a ship target is separated into two parts located at two adjacent subimages. As shown in Fig. 3, the overlap size is denoted by L.

2) Calculation of Densities: Then, we introduce the patch-based density features of the subimages to discriminate between the subimages containing pure clutter and those containing sparse targets. The density [27] ρ_i of the ith patch

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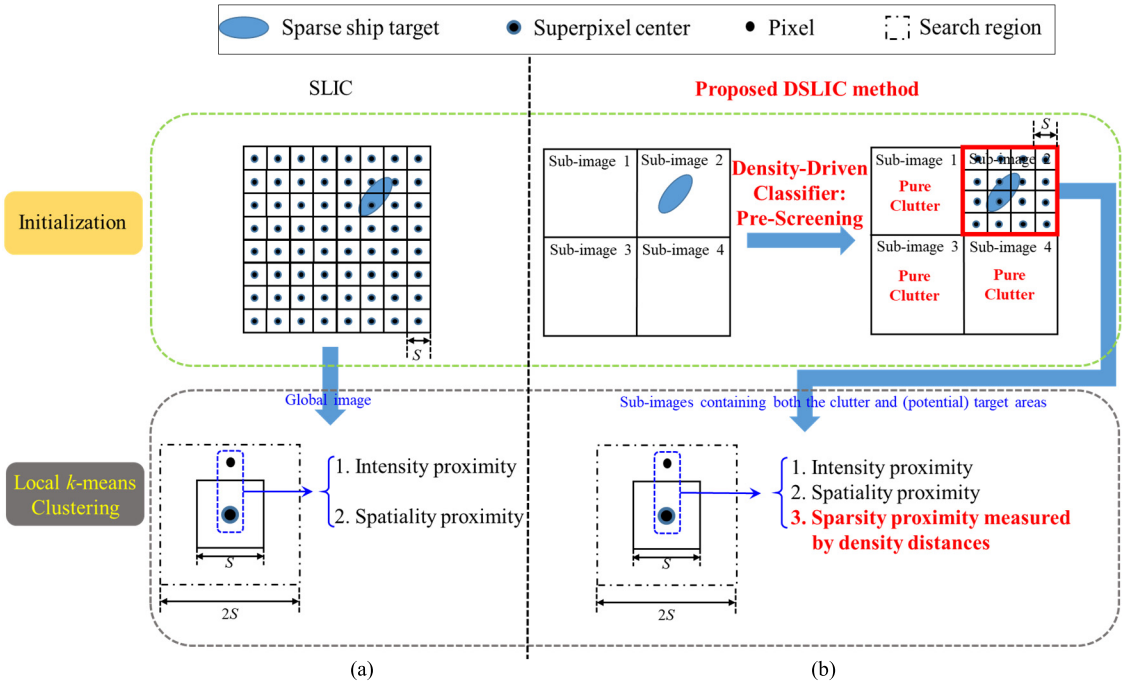


Fig. 2. Comparison of (a) SLIC [7] and (b) proposed DSLIC method for superpixel-based segmentation of marine SAR images. In DSLIC, a new density-driven classifier discards most of the subimages with pure clutter, i.e., it makes the segmentation process more efficient. The concept of sparsity proximity newly considered in our DSLIC enriches the representation of pixel proximity.

with $P_1 \times P_2$ (e.g., 3×3) pixels in a subimage represents the sparsity degree of the current patch in the whole subimage space

$$\rho_i = \sum_{j=1,2,\ldots,N^{(\text{patc})}, j\neq i} \text{Indi}\left(\bar{d}_{i,j}^{(\text{inte})} - d^{(\text{c})}\right) \tag{2}$$

where $\bar{d}_{i,j}^{(\text{inte})} = \left|\bar{\delta}_i - \bar{\delta}_j\right|$ denotes the distance between the average intensities $\{\bar{\delta}_i, \bar{\delta}_j\}$ of the ith patch and the jth patch for $i, j = 1, 2, \ldots, N^{(\text{patc})}$, $N^{(\text{patc})} = \lceil M_1 M_2/(P_1 P_2)\rceil$ is the number of patches in a subimage, $M_1 > P_1$, $M_2 > P_2$, $\lceil\cdot\rceil$ denotes the ceiling function, $\text{Indi}(x)$

$$\text{Indi}(x) = \begin{cases} 1, & \text{if } x < 0 \\ 0, & \text{otherwise} \end{cases}$$

denotes the indicator function, and $d^{(\text{c})}$ is a constant representing the cutoff distance satisfying the condition $0 < d^{(\text{c})} < \max_{i,j} \bar{d}_{i,j}^{(\text{inte})}$. Instead of pixel-based densities, patch-based densities are calculated in (2) because they are more robust to speckle noise in marine SAR images. In addition, the complexity of the calculation of patch-based densities in a subimage is $O\{M_1 M_2 + [N^{(\text{patc})}]^2\}$ that is smaller than that of the pixel-based version, i.e., $O\{[M_1 M_2]^2\}$, when $P_1, P_2 > 1$.

From the definition in (2), we have the following two observations.

1) If a subimage only contains sea clutter patches, then all the values in the density map of the current subimage are highly similar. This is because, for each clutter patch, there are always lots of other clutter patches similar to it in the whole subimage space.
2) On the contrary, if there are some sparse target patches in a subimage, their densities are significantly lower than those of other nonsparse clutter patches in the current subimage since the number of patches (in the whole subimage space) that are similar to a target patch is far smaller than those that are similar to a clutter patch.

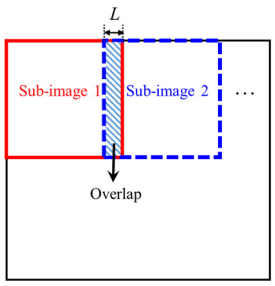


Fig. 3. Sketch of the overlap between two adjacent subimages in a large marine SAR image.

The two aforementioned characteristics can also be observed in Fig. 4, where a subimage with pure clutter and a subimage with target pixels can be well-differentiated by the density features in (2).

3) Density-Driven Classifier: Next, we design a density-driven classifier to prescreen each subimage and identify pure clutter ones and those containing targets, using the aforementioned density differences between them.

A very simple unsupervised classifier is the thresholding operation in which, if the minimum value of patch-based densities in a subimage is smaller than a predefined decision threshold, then this subimage is regarded as a candidate one (potentially containing ship targets) and will be preserved; otherwise, it is removed and will not be segmented. However,

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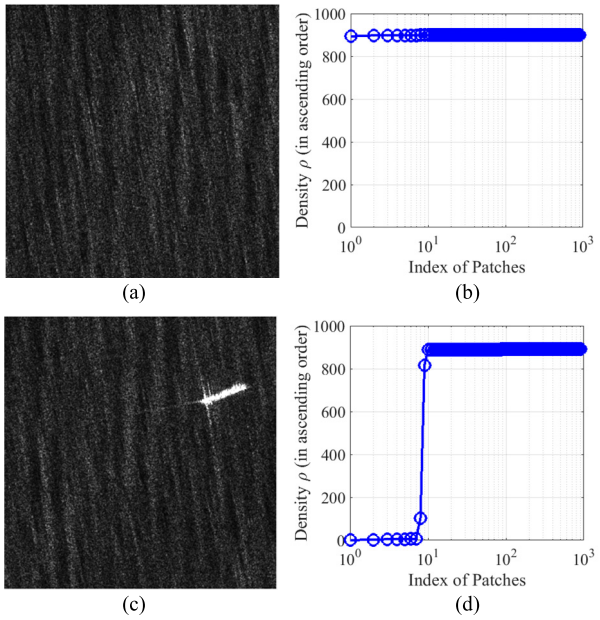


Fig. 4. (a) Subimage (300 × 300 pixels) with pure sea clutter. (b) Densities of patches in (a) in ascending order. (c) Subimage (300 × 300 pixels) with a sparse ship target. (d) Densities of patches in (c) in ascending order. Here, the patch size is $P_1 = P_2 = 10$, and $d^{(c)} = 0.2 \times \max_{i,j} \bar{d}_{i,j}^{(inte)}$, $i, j = 1, 2, \ldots, N^{(patc)}$ [for the calculation of density in (2)]. The resolution of SAR images is 3 m × 3 m.

it is difficult to determine an optimal decision threshold to avoid the case that a subimage with targets is misclassified into a pure clutter one. This misclassification directly leads to the loss of targets since target superpixels in this subimage will be unavailable. Existing deep neural network (DNN)-based classifiers [34], [35] perform a supervised classification framework relying on the availability of labeled data and often achieve superior classification performance. It is worth emphasizing that the operations in superpixel-based segmentation (and its initialization stage) should be computationally simple [7], while DNN-based classifiers contain lots of network layers and nonlinear activation functions, increasing the computation burden of superpixel-based segmentation, especially in onboard satellites. Due to the aforementioned reasons, we decided to design a simple but effective linear classifier (trained via the WSLMS strategy) to prescreen each subimage by exploiting its density features.

Let $\boldsymbol{\rho} \in \mathbb{R}^{N^{(patc)} \times 1}$ denote the vector containing all the patch-based densities in a test subimage, where densities are sorted in ascending order ($\rho_1 \leq \rho_2 \leq \cdots \leq \rho_{N^{(patc)}}$). The density-driven linear classifier for $\boldsymbol{\rho}$ is

$$y = \text{sign}\left(\tilde{\mathbf{w}}^T \boldsymbol{\rho} + \tilde{b}\right) \tag{3}$$

$$\begin{cases} \text{Removing:} \\ \text{Sub-image with the Pure Clutter,} \quad \text{if } y = -1 \\ \text{Perserving:} \\ \text{Sub-image with Target and Clutter Areas,} \quad \text{if } y = 1 \end{cases} \tag{4}$$

where $\tilde{\mathbf{w}} \in \mathbb{R}^{N^{(patc)} \times 1}$ is the weight vector, $\tilde{b} \in \mathbb{R}$ denotes the bias, sign(.) is 1 if the input element is positive and −1 otherwise, and y is the output of the linear classifier. The weight vector $\tilde{\mathbf{w}}$ and the bias $\tilde{b}$ are obtained before conducting the superpixel-based segmentation task via the following WSLMS mechanism:

$$\begin{aligned} \left\{\tilde{\mathbf{w}}, \tilde{b}\right\} &= \min_{\mathbf{w},b} \left\| \boldsymbol{\Lambda}\left(\check{\boldsymbol{\Theta}}\mathbf{w} + \mathbf{b} - \check{\mathbf{y}}\right)\right\|_2, \quad \text{s.t. } \|\mathbf{w}\|_0 \leq K \\ \check{\boldsymbol{\Theta}} &\triangleq \left[\check{\boldsymbol{\rho}}_1, \ldots, \check{\boldsymbol{\rho}}_q, \ldots, \check{\boldsymbol{\rho}}_Q\right]^T \in \mathbb{R}^{Q \times N^{(patc)}} \\ \check{\mathbf{y}} &\triangleq \left[\check{y}_1, \ldots, \check{y}_q, \ldots, \check{y}_Q\right]^T \in \mathbb{R}^{Q \times 1} \end{aligned} \tag{5}$$

where $\mathbf{b} = b \times \mathbf{1} \in \mathbb{R}^{Q \times 1}$, $\check{\boldsymbol{\rho}}_q \in \mathbb{R}^{N^{(patc)} \times 1}$, and $\check{y}_q$ are the sorted patch-based density vector (in ascending order) and the corresponding label (with/without target pixel) of the qth training subimage, respectively, Q is the number of training subimages, $\boldsymbol{\Lambda} \in \mathbb{R}^{Q \times Q}$ is a diagonal matrix, and K is the upper bound of the number of nonzero elements in $\mathbf{w}$ satisfying $K \ll N^{(patc)} < M_1 M_2$. The qth diagonal element in $\boldsymbol{\Lambda}$ is defined by

$$\lambda_q \triangleq \exp\left(\kappa_q\right) \tag{6}$$

where κ_q is the number of ship targets in the qth training subimage for $q = 1, 2, \ldots, Q$. The optimization problem in (5) can be solved using training data via existing compressed sensing methods [37], [38]. Here, the orthogonal matching pursuit (OMP) algorithm in [37] is selected to solve (5) because of its superior performance in terms of sparse signal recovery.

There are two main characteristics of the WSLMS in (5) are given as follows.

1) The L_0 constraint enhances the sparsity of $\tilde{\mathbf{w}}$. The discrimination between subimages with pure clutter and those with target objects mainly relies on a few elements with relatively small values in $\check{\boldsymbol{\rho}}_q$, as observed from Fig. 4(b) and (d). Accordingly, only a small number of elements in the weight vector $\tilde{\mathbf{w}}$ should be active (i.e., nonzero values). In addition, the L_0 constraint in (5) is helpful to fast screen subimages since the complexity of the classifier in (3) is only $O(K)$.
2) The weighted loss characterized by $\boldsymbol{\Lambda}$ promotes a large loss when a subimage with ship targets is misclassified into a pure clutter one. Furthermore, the cost function in (5) is increasing, with a larger number of targets in the current subimage. This is beneficial to preserve all the subimages with targets for the subsequent local clustering stage at the expense of potentially preserving a small number of "false alarm" subimages (with pure clutter) in a large marine SAR image. It is worth mentioning that these potential "false alarm" subimages will also be segmented in the subsequent local clustering stage of DSLIC but do not imply real false alarm areas in target detection results (after the superpixel-based segmentation).

The proposed prescreening operation is performed before the local k-means clustering operation to achieve sparse initialization of superpixel centers. A few prescreening methods have been proposed in the context of SAR ship detection, e.g., global constant false alarm rate (CFAR) [26], [62], [64] and maximum stable extremum region (MSER) [63]. Note that

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SAR ship detection aims to obtain a higher detection rate with a fixed false alarm rate, while superpixel-based segmentation is a preprocessing stage of SAR ship detection to maintain the compactness of superpixels with a high computation efficiency. Existing methods based on global CFAR and MSER do not consider the discrimination between ship targets and sea clutter in terms of sparsity level in spatiality. They show limited prescreening performance since the nonsparse sea clutter areas may be preserved. In this article, the density features of marine SAR images are innovatively exploited to measure the sparsity level of pixels and achieve the prescreening of SAR images. The exploiting of density features enhances the identification performance of nonsparse sea clutter. In Section V, we will compare the performance of global CFAR [26], MSER [63], and our prescreening operation using real SAR images. We also notice that, in [35] and [65], a prescreening operation of SAR images (according to the bias classification strategy) is applied to remove land clutter and accelerate the subsequent deep-learning-based SAR ship detection method. Our work is different from [35] and [65] since the initialization in DSLIC aims at removing sea clutter in accordance with the density features in SAR images.

B. Stage 2: Local K-Means Clustering by Aggregating Density Features

In this section, we describe how to process each subimage [preserved by the density-based classifier in (3)] using a local k-means clustering strategy. The proximity measure in (1) does not consider the difference between the sparsity degrees of two pixels, leading to the potential coexistence of sparse target areas and nonsparse clutter areas within superpixels. Therefore, in addition to the spatiality proximity and the intensity proximity, the sparsity proximity based on density distances is embedded into the proximity term in (1). For the ith pixel and the jth pixel in a subimage, the proposed new proximity measure is

$$D(i,j)=\sqrt{\left(\frac{d_{i,j}^{(\mathrm{spat})}}{\tau_1}\right)^2+\left(\frac{d_{i,j}^{(\mathrm{inte})}}{\tau_2}\right)^2+\left(\frac{d_{i,j}^{(\mathrm{spas})}}{\tau_3}\right)^2} \tag{7}$$

where

$$d_{i,j}^{(\mathrm{spas})}=\left|\rho_i-\rho_j\right| \tag{8}$$

represents the sparsity proximity and τ_3 is the corresponding normalized factor. Pixel-based densities $\{\rho_i,\rho_j\}$ in (8) are degenerated versions of the patch-based densities in (2) with $P_1=P_2=1$. The newly added sparsity proximity term $d_{i,j}^{(\mathrm{spas})}$ in (7) is helpful to cluster sparse target pixels into target superpixels and exclude nonsparse sea clutter pixels therein. The superpixel-based segmentation results for the current subimage can be obtained by using the local k-means clustering with the proximity measure in (7).

Note that 3×3 Gaussian kernel filtering [21] is performed on the preserved subimages before the calculation of (7). This is to reduce the effect of speckle noise on both the intensity proximity and sparsity proximity. The normalized factors τ_1,τ_2,τ_3 are adaptively selected using the maximum values of $d_{i,j}^{(\mathrm{spat})}$, $d_{i,j}^{(\mathrm{inte})}$, and $d_{i,j}^{(\mathrm{spas})}$ for each superpixel center, respectively, from the previous iteration of the local k-means clustering [7].

Algorithm 1 DSLIC

Input: Marine SAR image with N pixels, superpixel size S, sub-image size $M_1\times M_2$, overlap size L, patch size $P_1\times P_2$, maximum number of iterations $T_{\max}$, weight vector $\tilde{\mathbf{w}}$, and bias $\tilde{b}$.

Stage 1: *Initialization Based on Density-Driven Classifier.*

1) Uniformly segment the input marine SAR image into several $M_1\times M_2$ sub-images with overlap size L.
2) Calculate patch-based density features in each sub-image based on (2).
3) Sort the patch-based densities within each sub-image in ascending order and pre-screen the sorted results via the density-driven classifier in (3) and (4).
4) For each sub-image preserved by (3) and (4), initialize superpixel centers in accordance with regular grids with the interval S. Then, move each superpixel center to the position with smallest gradient value in its 8-neighbor connectivity region.

Stage 2: *Local K-Means Clustering by Aggregating Density Features.*

For each sub-image preserved by (3) and (4):
5) Associate each pixel to its nearest superpixel center according to the proximity in (7), where the search area is $2S\times 2S$ -pixels around each superpixel center.
6) Update the position of each superpixel center using the mean coordinate of the pixels associated with it.
7) Repeat steps 5) and 6) until the number of iterations exceeds a predefined value $T_{\max}$.
8) Perform postprocessing to enforce connectivity: "orphan pixels" are associated with its nearest superpixel, as in [7].

Output: Superpixels in the marine SAR image.

A detailed implementation of the proposed DSLIC superpixel-based segmentation approach is summarized in Algorithm 1.

IV. Computational Complexity and Memory Requirements of DSLIC

In this section, we analyze the computational complexity and memory requirements of the proposed DSLIC method. In order to simplify our analysis, we assume that the overlap of adjacent subimages in Stage 1 of DSLIC is negligible since a small size of the overlap will not dramatically increase the number of subimages in an SAR image. Therefore, let us assume that the number of $M_1\times M_2$ subimages in a large marine SAR image with N pixels is $N^{(\mathrm{subimg})}\triangleq\lceil N/(M_1M_2)\rceil$. Let $\varepsilon\triangleq N^{(\mathrm{preser})}/N^{(\mathrm{subimg})}$ represent the "preserving ratio," where $N^{(\mathrm{preser})}$ is the number of preserved subimages after the prescreening conducted via the density-driven classifier in (3).

In Stage 1 of DSLIC, the complexity of multiplication and addition operations required by the calculation of patch-based densities is $O(N^{(\mathrm{subimg})}N^{(\mathrm{patc})})$ and $O\{N^{(\mathrm{subimg})}[(N^{(\mathrm{patc})})^2+M_1M_2]\}$, respectively. The additions in the sorting operation related to densities have $O(N^{(\mathrm{subimg})}N^{(\mathrm{patc})}\log_2 N^{(\mathrm{patc})})$ complexity. The complexity of the multiplication and addition operations involved in the density-driven classifier in (3) is $O(N^{(\mathrm{subimg})}K)$. In Stage 2 of DSLIC, the Gaussian kernel filtering on each preserved subimage involves multiplications/additions with $O(M_1M_2)$ complexity. The complexity of

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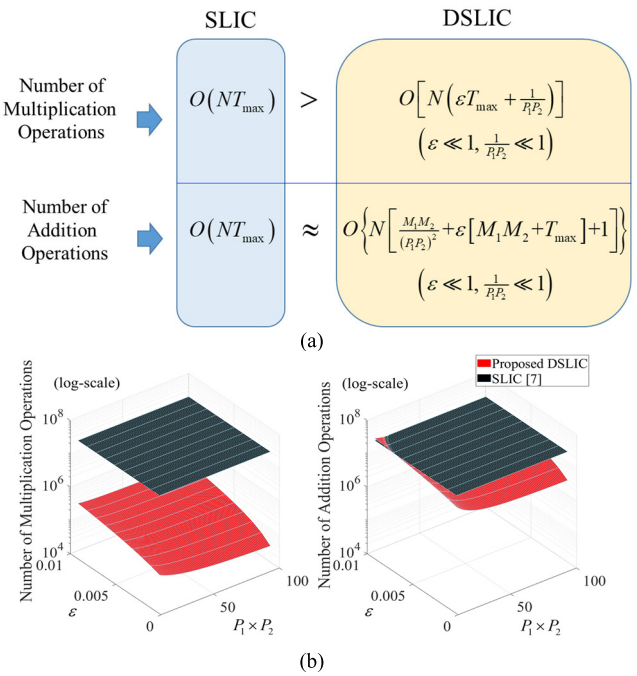


Fig. 5. (a) Mathematical comparison of SLIC [7] and the proposed DSLIC method in terms of computational requirements. (b) Numerical results associated with (a), where $N = 10^6$, $T_{\max} = 20$, and $M_1 \times M_2 = 1600$.

the multiplication and addition operations for the calculation of proximity in (7) is $O(T_{\max} M_1 M_2)$. The complexity of the addition operations required by the calculation of pixel-based densities in (8) is $O[(M_1 M_2)^2]$. Therefore, the overall complexity of the multiplication and addition operations in DSLIC is

$$\begin{aligned}\text{Multiplication:}&O(N^{(\text{subimg})} N^{(\text{patc})} + N^{(\text{subimg})} K \\ &+ T_{\max} M_1 M_2 N^{(\text{preser})} + M_1 M_2 N^{(\text{preser})}) \quad (9)\\ \text{Addition:}&O\{N^{(\text{subimg})}[(N^{(\text{patc})})^2 + M_1 M_2 + N^{(\text{patc})} \\ &\times \log_2 N^{(\text{patc})} + K] + N^{(\text{preser})} M_1 M_2 \times (M_1 M_2 + T_{\max} + 1)\} \quad (10)\end{aligned}$$

respectively. Based on $T_{\max} > 1$, $N^{(\text{patc})} \gg K$, $N^{(\text{patc})} > \log_2 N^{(\text{patc})}$, and the approximations of $N^{(\text{subimg})} = \lceil (N/M_1 M_2) \rceil \approx (N/M_1 M_2)$ and $N^{(\text{patc})} = \lceil (M_1 M_2 / P_1 P_2) \rceil \approx (M_1 M_2 / P_1 P_2)$, the complexity of multiplication and addition operations required by DSLIC in (9) and (10) can be further simplified to

$$\text{Multiplication:} O\left[N\left(\varepsilon T_{\max} + \frac{1}{P_1 P_2}\right)\right] \quad (11)$$

$$\text{Addition:} O\left\{N\left[\frac{M_1 M_2}{(P_1 P_2)^2} + \varepsilon(M_1 M_2 + T_{\max}) + 1\right]\right\} \quad (12)$$

respectively.

The computational requirements of SLIC and the proposed DSLIC method are summarized in Fig. 5 in terms of mathematical and numerical aspects. As shown in Fig. 5, the number of multiplication operations in DSLIC is dramatically smaller than that in SLIC when $\varepsilon \ll 1, (1/P_1 P_2) \ll 1$. This benefits from the fact that the local k-means clustering stage in DSLIC only needs to process a small number of subimages, as opposed to the global SAR image in SLIC. Note that the condition $(1/P_1 P_2) \ll 1$ is easy to be satisfied (e.g., $P_1, P_2 \geq 3$). In this work, we mainly focus on offshore scenarios, where ship targets occupy far fewer pixels than sea clutter and ship targets do not often appear in each subimage of a large marine SAR image (covering offshore areas with tens/hundreds of square kilometers). Thus, the condition of $\varepsilon \ll 1$ is often satisfied. The calculation of densities in DSLIC mainly requires addition operations. However, the total number of addition operations in DSLIC $O\{N \times [(M_1 M_2/(P_1 P_2)^2) + \varepsilon(M_1 M_2 + T_{\max}) + 1]\}$ may be comparable to that of SLIC $O(N \times T_{\max})$. For example, let $T_{\max} = 20$, $P_1 = P_2 = 6$, and $M_1 = M_2 = 40$, and assume $\varepsilon = 0.01$. Then, we have $(M_1 M_2/(P_1 P_2)^2) + \varepsilon(M_1 M_2 + T_{\max}) + 1 \approx 18$ that is comparable to $T_{\max} = 20$. Based on the above analysis, we conclude that the proposed DSLIC method requires much less computation than SLIC when processing marine SAR images. The specific running times and numbers of floating-point operations (FLOPs) required by the proposed DSLIC and SLIC [7] will be provided in Section V.

In the local clustering stage, besides a few scalar parameters, DSLIC needs to store $3\varepsilon N$ floating-point numbers, i.e., εN pixel intensities, εN pixel densities, and εN proximity measures [see $D(i, j)$ in (7)]. The memory requirement of SLIC [7] is $2N$, i.e., N pixel intensities and N proximity measures. Accordingly, we have that DSLIC requires much fewer memory resources in comparison with SLIC [7] in the case of $\varepsilon < 2/3$, which is easy to be satisfied by sparsely distributed ship targets in SAR images.

TABLE I
KEY PARAMETERS OF THE MARINE SAR IMAGE DATASET USED FOR EXPERIMENTS

	Figs. 6, 10, 13, and 15	Fig. 16
Satellite	Gaofen-3	Sentinel-1
Polarization	HH	VH
Frequency Band	C	C
Centre Frequency	5.4 GHz	5.405 GHz
Resolution (range×azimuth)	3m×3m	20m×22m

V. EXPERIMENTAL RESULTS

In this section, the superpixel-based segmentation performance and the complexity of the MST [17], SLIC [7], exponential-SLIC (E-SLIC) [24], pixel intensity and the location similarity (PILS) [21], and the proposed DSLIC method are compared using C-band Gaofen-3 and Sentinel-1 datasets. Then, the superpixels generated by the above five approaches are used for the task of superpixel-based ship detection to evaluate the segmentation performance in depth. The ground-truth locations of ship targets in marine SAR images are obtained from [32], [33], and [42], and the boundaries of ship targets are empirically human-labeled. In Table I, we list the key parameters of the marine SAR images that are used in this section. Note that the accurate incidence angle of each SAR image in this section is not available in open

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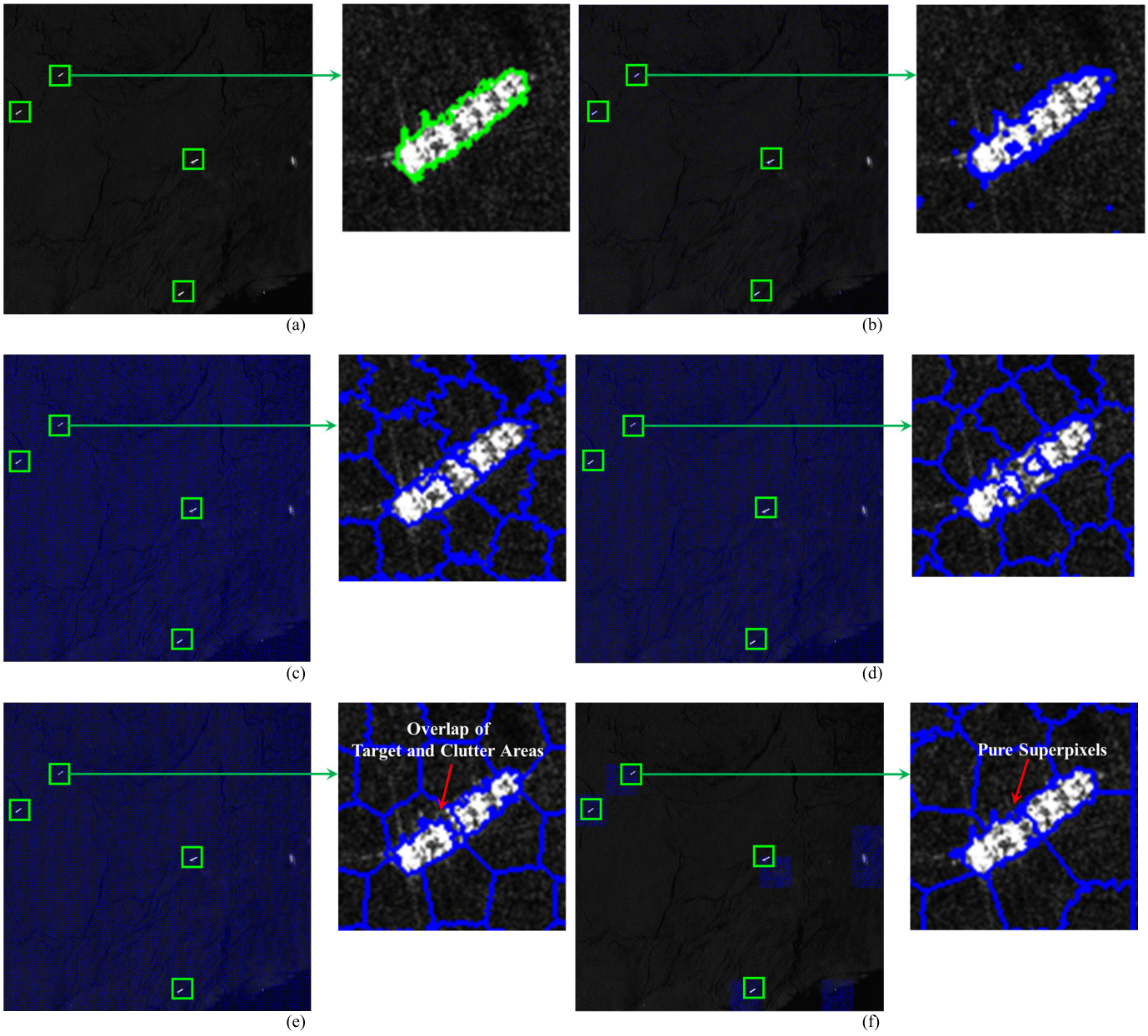


Fig. 6. (a) Gaofen-3 SAR image (3000 × 3000 pixels) with heterogeneous sea clutter. The sea state code of (a) is 4 that is the largest one in the dataset [42]. Superpixel-based segmentation results of (b) MST [17], (c) PILS [21], (d) E-SLIC [24], (e) SLIC [7], and (f) proposed DSLIC, where $S = 22$, $T_{\max} = 10$. Green lines in the subimage of (a) denote the ground-truth boundaries of the ship target. Blue lines in (b)–(f) represent the boundaries of superpixels. Green rectangles are used to identify the location of ship targets.

datasets [32], [33], [42]. In a way, the effect of the incidence angles can be reflected by the quality of SAR images. In this section, the experimental results based on SAR images with significantly different quality will imply the performance of our proposed method with different incidence angles. Similar to existing superpixel-based marine SAR image processing methods in [10], [23], [25], [26], [40], and [50], we do not consider the effect of ship motions. We refer readers to [59] and [60] for the image features based on ship motions and autofocusing methods for moving ship targets in SAR images. Experiments are performed using MATLAB (2021a) in a 64-bit Windows operating system with 1 TB of RAM and an Intel Xeon CPU. The uncompressed versions and ground truth of Figs. 6, 10, and 16 can be found in the supporting files associated with this article (see the Supplementary Material).

Inspired by the image chip sizes in open SAR datasets [33], [41], we choose $M_1 = M_2 = 300$ for DSLIC. The resolutions of marine SAR images influence the choice of the hyperparameter of superpixel size S. We adopt the strategy used in [10] for determining the superpixel size S, where S is empirically selected based on a percentage (e.g., 25%) of the average number of pixels in the region occupied by a ship target. This setting captures the shape of ship targets well in high-resolution SAR images. In addition, the superpixel size S in this article ranges from 12 to 30, similar to existing works regarding superpixel-based marine

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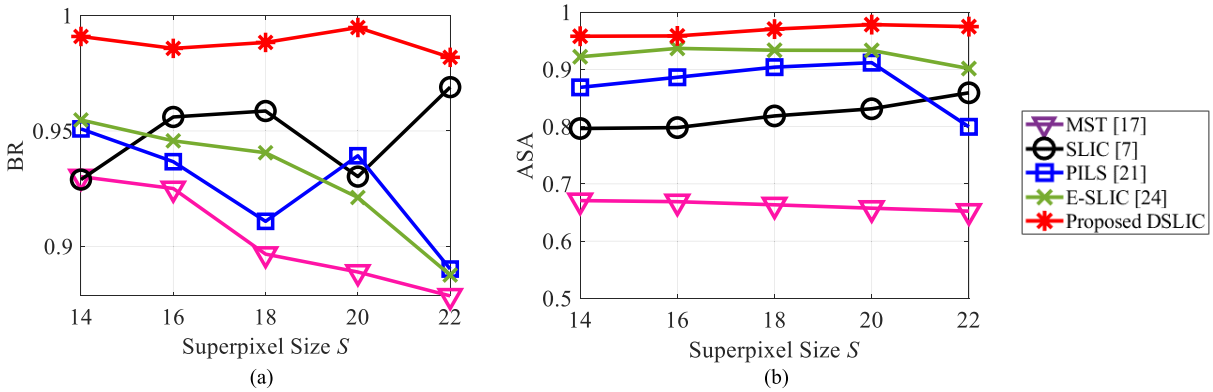


Fig. 7. BR and ASA curves of the segmentation results in Fig. 6 with different superpixel sizes. (a) BR values. (b) ASA values.

SAR image processing in [10] and [12]. The above settings of M_1, M_2, and S also satisfy $M_1 \times M_2 \gg S^2$. Without loss of generality, we set $P_1 = P_2 = 10$ for Figs. 6 and 10 (with 3 m $\times$ 3 m resolution) and $P_1 = P_2 = 5$ for Fig. 16 (with 20 m $\times$ 22 m resolution) to reduce the effect of speckle noise and accelerate Stage 1 in DSLIC. The overlap size is $L = 0$. We choose $L = 0$ just because it is convenient to quantitatively evaluate the segmentation performance of DSLIC and the corresponding detection performance when there is no overlap between two adjacent subimages. In the case of $L > 0$, the nonmaxima suppression (NMS) operation [49] may be used to refine superpixels in the overlap regions. The cutoff distance $d^{(c)}$ in (2) is adaptively selected as

$$d^{(c)} = \beta \times \max_{i,j} \bar{d}_{i,j}^{(\text{inte})} \tag{13}$$

where $\beta = 0.2$. The robustness of DSLIC on the selection of the patch size $P_1 \times P_2$, the cutoff distance $d^{(c)}$, and the superpixel size S will be investigated in Sections V-A and V-B. The weight vector $\tilde{\mathbf{w}}$ and bias $\tilde{b}$ in (5) are trained based on 60 SAR image chips with pure clutter and 60 SAR image chips containing ship targets, where $K = 4$. The numbers of training subimages with one to seven ship targets are 40, 10, 4, 3, 1, 1, and 1, respectively.

A. Case Study: Marine SAR Image in Offshore Scenario With Heterogeneous Sea Clutter

1) Visualization of Superpixels: Fig. 6 depicts the superpixels of a marine SAR image with heterogeneous sea clutter in an offshore scenario, where green lines represent the ground-truth boundaries of ship targets, blue lines denote the boundaries of generated superpixels, and the green rectangles are used to identify the location of ship targets. The superpixel size is $S = 22$. Heterogeneous sea clutter means that there are different sea clutter levels in Fig. 6(a) caused by sea currents or winds. Note that MST [17] requires a predefined number of superpixels as an input parameter instead of the superpixel size. Here, the predefined number of superpixels for MST [17] is $\lceil N/S^2 \rceil$. As shown in Fig. 6(b), MST [17] successfully segments ship targets and retains the integrity of sea clutter areas to a great extent. However, MST [17] suffers from some small clutter superpixels. This is because MST [17] aims at producing multiscale superpixels and may be affected by the speckle noise in SAR images. SLIC-based methods [7], [21], [24] show better speckle suppression than MST [17]. In Fig. 6(c) and (d), PILS [21] and E-SLIC [24] provide similar segmentation performance with irregular superpixels. In Fig. 6(e) and (f), the superpixels generated by SLIC [7] and the proposed DSLIC method are regular. It is worth emphasizing that there is an overlap of target and clutter areas within superpixels generated by SLIC [7], while DSLIC provides more internally compact superpixels than SLIC. This demonstrates the superiority of the concept of sparsity proximity newly considered in the local clustering stage of DSLIC. Besides, DSLIC provides elaborate superpixels only in seven subimages (in the original large marine SAR image). The preserving ratio is $\varepsilon = N^{(\text{perser})}/N^{(\text{subimg})} = 7/100 = 0.07 \ll 1$. These seven subimages cover all the four ship targets in Fig. 6(a). This implies the effectiveness of the weight vector $\tilde{\mathbf{w}}$ and the bias $\tilde{b}$, estimated by the WSLMS strategy in (5).

2) Quantitative Metrics of Segmentation Performance: In Fig. 7, the segmentation results visually displayed in Fig. 6 are quantitatively evaluated by using two commonly used metrics in the literature, i.e., boundary recall (BR) and achievable segmentation accuracy (ASA) belonging to [0, 1]. Larger values of BR and ASA mean that the generated superpixels exhibit better boundary consistency and internal compactness, respectively. We refer readers to [44] for detailed definitions of BR and ASA. It is worth mentioning that a small segmentation error can lead to perfect BR = 1 [44] since the definition of BR considers the fact that the human-labeled ground truth of ship targets in SAR images may contain slight deviations. As illustrated in Fig. 7, the proposed DSLIC method outperforms PILS [21], MST [17], SLIC [7], and E-SLIC [24] in terms of both BR and ASA metrics. This is because the sparsity proximity concept in DSLIC reduces the overlap of sparse target areas and nonsparse clutter areas within superpixels.

3) Robustness to Cutoff Distance: In Fig. 8, we investigate the segmentation performance of the proposed DSLIC method versus the cutoff distance $d^{(c)}$ in (2) determined by the scale factor β in (13) using the SAR image in Fig. 6(a). As shown in Fig. 8, the proposed DSLIC algorithm provides similar values of BR and ASA with different cutoff distances and

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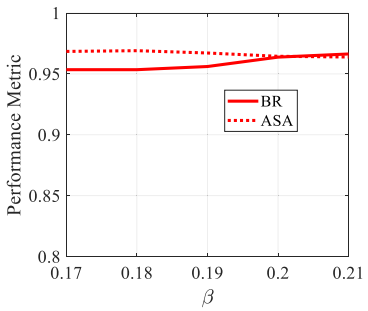


Fig. 8. Superpixel segmentation performance of the proposed DSLIC method versus the scale factor β to determine the cutoff distance $d^{(c)}$ in (13), where $S = 22$. Here, the SAR image in Fig. 6(a) is used.

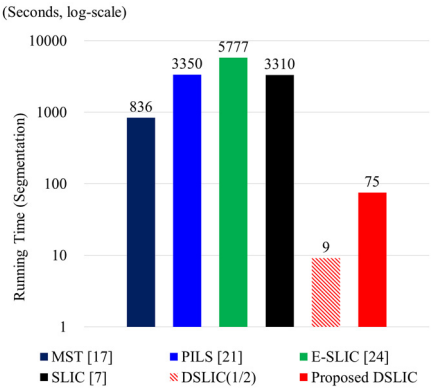


Fig. 9. Comparison of the running times of different superpixel segmentation methods, where the SAR image in Fig. 6(a) is used and the superpixel size is $S = 30$.

is still competitive in comparison with other methods (see the comparison between Figs. 7 and 8).

4) Specific Segmentation Time and FLOPs: In Fig. 9, we compare the total running times of MST [17], SLIC [7], E-SLIC [24], PILS [21], DSLIC (1/2), and the proposed DSLIC method. The initialization stage of DSLIC (1/2) is the same as that in DSLIC, but the sparsity proximity in the local clustering stage is not considered. From Fig. 9, we can see that SLIC [7] and its state-of-the-art extensions [21], [24] require more computation cost than MST [17]. This is because MST [17] is designed based on the Boruvka algorithm without the iterative update of superpixel centers that are required by SLIC [7] and its extensions [21], [24]. The proposed DSLIC method only requires 1.3%–9.0% running time compared with other methods in [7], [21], [17], and [24] since DSLIC does not finely segment most of the noninformative clutter areas. It is worth mentioning that the fastest segmentation performance is attained by DSLIC (1/2), where densities of pixels are not required to be calculated in the local clustering stage (such as SLIC). DSLIC (1/2) can be regarded as a rapid implementation version of DSLIC (at the expense of segmentation performance loss because the segmentation performance of DSLIC (1/2) is approximately the same as that of SLIC). In Table II, we count the numbers of FLOPs when segmenting Fig. 6(a). The numbers of FLOPs required by different segmentation algorithms are inferred using the codes in [45]. Note that

TABLE II
NUMBER OF FLOPS WHEN PROCESSING FIG. 6(a)

Methods	Number of FLOPs
MST [17]	9.0×10^{8}
PILS [21]	2.6×10^{9}
E-SLIC [24]	6.0×10^{10}
SLIC [7]	7.4×10^{9}
Proposed DSLIC	3.2×10^{8}

TABLE III
AUC VALUES AND RUNNING TIMES OF THE SLCM DETECTOR [40] USING SUPERPIXELS GENERATED BY METHODS IN [7], [17], [21], AND [24] AND PROPOSED DSLIC METHOD. THE SAR IMAGE IN FIG. 6(a) IS USED, AND THE SUPERPIXEL SIZE IS $S = 18$

Methods	AUC	Running Time (Detection)
MST [17]	0.8356	659
PILS [21]	0.9255	638
E-SLIC [24]	0.9735	667
SLIC [7]	0.9251	711
Proposed DSLIC	0.9987	1

we have not found an effective way so far to count the number of addition and multiplication operations, respectively, in MATLAB code files. Therefore, we only count the total number of FLOPs for each algorithm in Table II. As shown in Table II, the proposed DSLIC requires the smallest number of FLOPs compared with other methods. The number of FLOPs required by DSLIC is only 4.3% of that required by SLIC [7]. This is consistent with the theoretical analysis in Section IV.

5) Robustness to Patch Size: The running times of DSLIC with 6×6, 8×8, and 10×10 patch sizes for the calculation of densities in (2) are 77, 55, and 45 s, respectively, where $S = 22$. We can see that the running times of DSLIC are still smaller than other existing methods with different patch sizes (see Fig. 9). Additional experiments also show that all the preserving ratios of the classifier in (3) are still $\varepsilon = 0.07$ with $P_1 = P_2 = 6$, 8, and 10, and the preserved subimages still cover all the target areas. The running times of DSLIC decrease with the increase in patch sizes, which is consistent with the theoretical result in Fig. 5.

6) Target Detection Performance and Complexity: In Table III, we show the area under the curve (AUC) values obtained by the SLCM detector [40] using the marine SAR image in Fig. 6(a). SLCM is developed based on the derived kernel model and is selected here owing to its simplicity and good detection performance, especially for weak ship targets [40]. The probability of detection (P_d) and the probability of false alarm (P_{fa}) involved by AUC are calculated as $P_d = N_{\text{dtar}}/N_{\text{alltar}}$ and $P_{\text{fa}} = N_{\text{dfal}}/N_{\text{allclu}}$, respectively, where N_{dtar} and N_{dfal} denote the numbers of detected target and false alarm pixels, respectively, and N_{alltar} and N_{allclu} are the numbers of all the target and clutter pixels in large marine SAR images, respectively. From Table III, we can see that the proposed DSLIC algorithm leads to higher AUC values, i.e., higher superpixel-based detection performance. This is due to two main advantages of DSLIC: 1) most of the clutter areas

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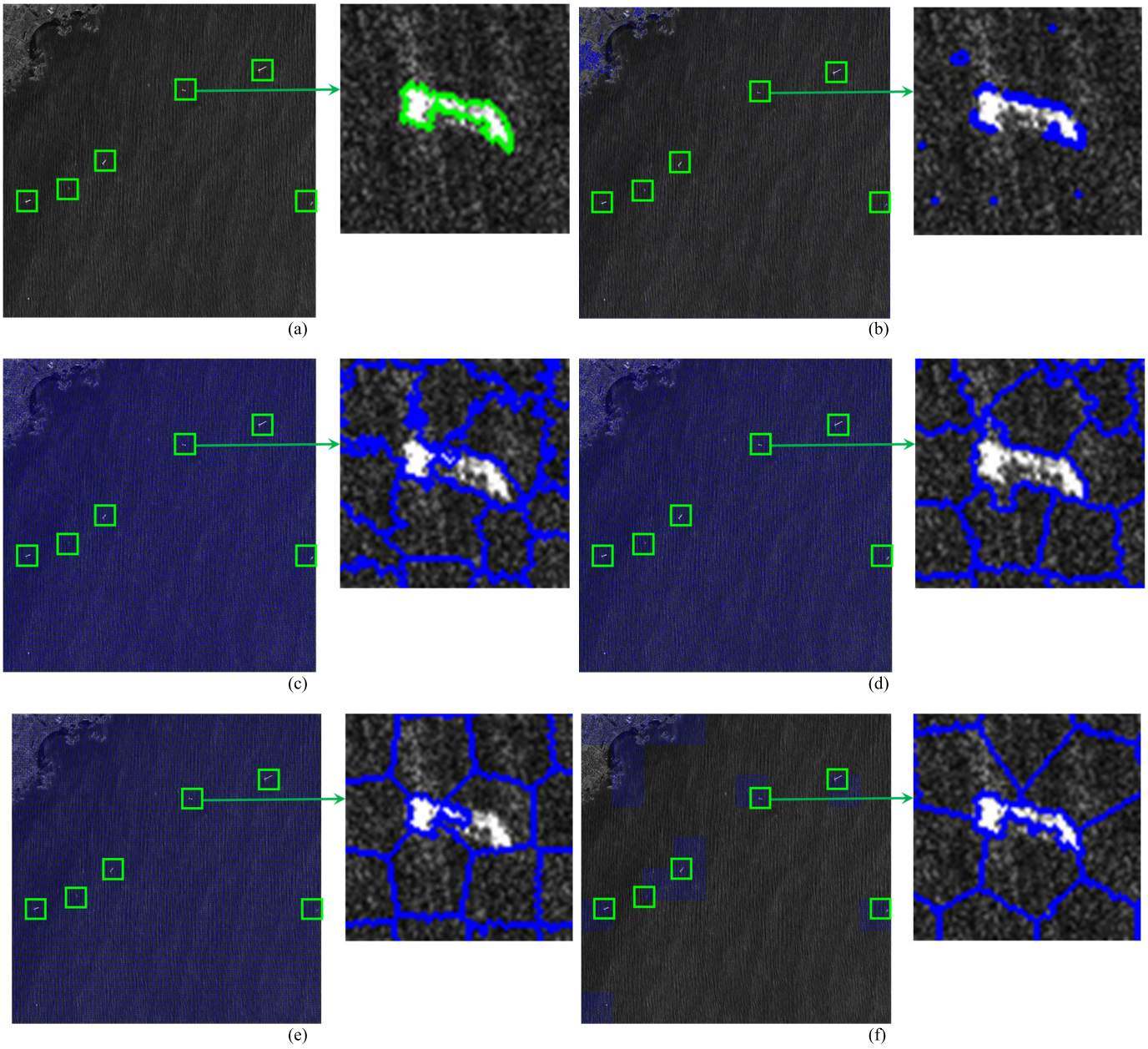


Fig. 10. (a) Gaofen-3 SAR image with low target-to-clutter contrast and land clutter pixels (3000 × 3000 pixels). The sea state code is 4. Superpixel-based segmentation results of (b) MST [17], (c) PILS [21], (d) E-SLIC [24], (e) SLIC [7], and (f) proposed DSLIC, where $S = 22$ and $T_{\max} = 10$. Green lines, blue lines, and green rectangles represent the ground-truth boundaries of true targets, boundaries of superpixels, and locations of ship targets, respectively.

(with low densities) are removed in the initialization stage of DSLIC and will not become false alarm areas in the detection results and 2) the newly embedded sparsity proximity concept (reflected by density distances) in DSLIC improves the compactness of target/clutter superpixels. In Table III, the specific running times of SLCM [40] are also provided using superpixels generated by MST [17], SLIC [7], E-SLIC [24], PILS [21], and the proposed DSLIC method. We can observe from Table III that the proposed DSLIC method dramatically reduces the computation burden of the subsequent detection task since only a small number of subimages with potential ship targets need to be elaborately segmented. This again demonstrates the superiority of the proposed DSLIC method in terms of computation efficiency.

B. Case Study: Marine SAR Image With Low Target-to-Clutter Contrast and Land Clutter

1) Visualization of Superpixels: In Fig. 10, we show superpixels of a marine SAR image with the low target-to-clutter contrast and a few land clutter pixels, where $S = 22$. From Fig. 10(b), we can see that MST [17] does not provide satisfactory segmentation performance, where weak ship targets are not circled well. In Fig. 10(c)–(e), the superpixels generated by PILS [21], E-SLIC [24], and SLIC [7] cannot completely capture the outline pixels of ship targets. The proposed DSLIC algorithm generates more compact superpixels in comparison with other methods. Similar to Fig. 6(f), it can be observed from Fig. 10(f) that the subimages preserved by

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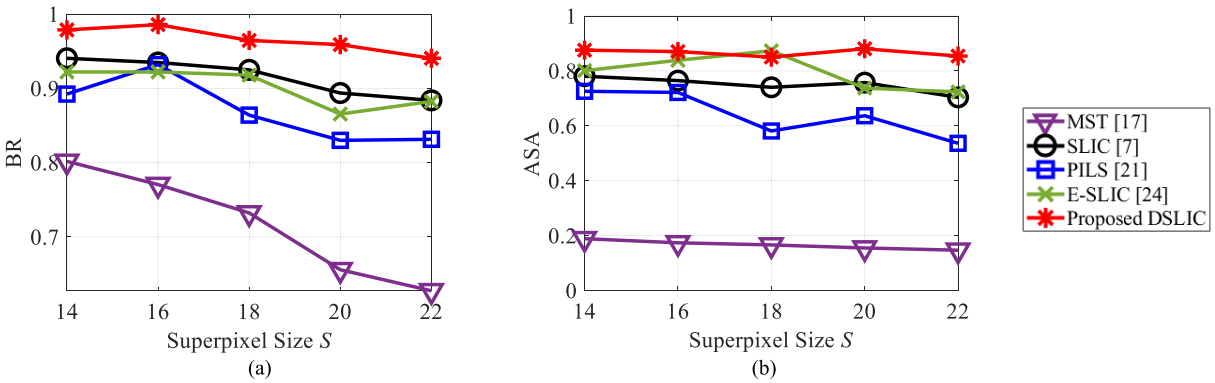


Fig. 11. BR and ASA curves of the segmentation results in Fig. 10 with different superpixel sizes. (a) BR values. (b) ASA values.

the initialization stage of DSLIC successfully find all the ship targets and remove most of the subimages with pure clutter. The preserving ratio of Fig. 10(f) is $\varepsilon = N^{(\text{perser})}/N^{(\text{subimg})} = 14/100 = 0.14 \ll 1$. It is worth mentioning that the complex land clutter in Fig. 10(a) increases the difficulty to identify whether a subimage contains ship targets. There are seven preserved subimages without targets in Fig. 10(f). The effect of land clutter can be reduced by using the existing sea-land segmentation algorithms [43] before the superpixel generation task. In Fig. 10(a), ship targets show different orientations. It is worth mentioning that the proposed DSLIC method is still effective with different geometric relations between the SAR sensors and ships due to the natural sparsity of ship targets in large SAR images.

2) Quantitative Metrics of Segmentation Performance: In Fig. 11, we plot the BR and ASA curves of DSLIC (and other competitors) versus the superpixel size using the SAR image in Fig. 10(a). We can observe from Fig. 11 that MST [17] exhibits smaller BR and ASA values since the number of pixels in a superpixel produced by MST is not restricted, leading to a loss of boundary information for small and weak ship targets. The proposed DSLIC approach and E-SLIC [24] perform similarly and exhibit better segmentation performance with higher BR and ASA values in comparison with MST [17], SLIC [7], and PILS [21].

3) Specific Segmentation Time: Fig. 12 shows the running times of different superpixel-based segmentation methods when processing Fig. 10(a). Similar to Fig. 9, DSLIC exhibits dramatically higher computation efficiency compared with other state-of-the-art methods in [7], [17], [21], and [24]. The running time of DSLIC when processing the image in Fig. 10(a) is larger than that when processing the image in Fig. 6(a) since more subimages are segmented in the local clustering stage for the former image. The numbers of FLOPs of aforementioned superpixel-based segmentation methods to process Fig. 10(a) are approximately the same as those in Table II and are not reported here.

4) Target Detection Performance and Complexity: In Table IV, the AUC values and running times of the SLCM detector [40] are presented using the image in Fig. 10(a), where the superpixel-based segmentation methods in [7], [17], [21], and [24] and the proposed DSLIC method are applied as the preprocessing step therein. Table IV reveals that DSLIC again results in better detection performance and less computation cost. After comparing Tables III and IV, we can find that the land clutter and strong sea clutter degrade the superpixel-based detection performance.

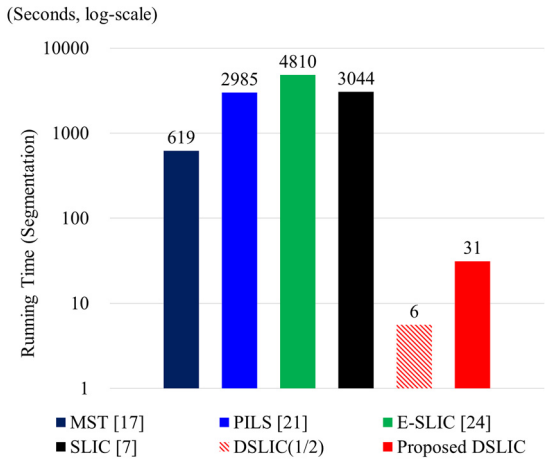


Fig. 12. Comparison of the running times of different superpixel-based segmentation methods, where the SAR image in Fig. 10(a) is used and the superpixel size is $S = 30$.

TABLE IV

AUC VALUES AND RUNNING TIMES OF THE SLCM DETECTOR [40] USING SUPERPIXELS GENERATED BY METHODS IN [7], [17], [21], AND [24] AND THE PROPOSED DSLIC METHOD. THE SAR IMAGE IN FIG. 10(a) IS USED, AND THE SUPERPIXEL SIZE IS $S = 18$

Methods	AUC	Running Time (Detection)
MST [17]	0.5861	680
PILS [21]	0.7793	758
E-SLIC [24]	0.9311	782
SLIC [7]	0.9124	665
Proposed DSLIC	0.9896	2

C. Case Study: Robustness of Prescreening Operation Versus Sea State Code

The quality of marine SAR images is dependent on the sea state [66]. In practice, a larger sea state code implies a

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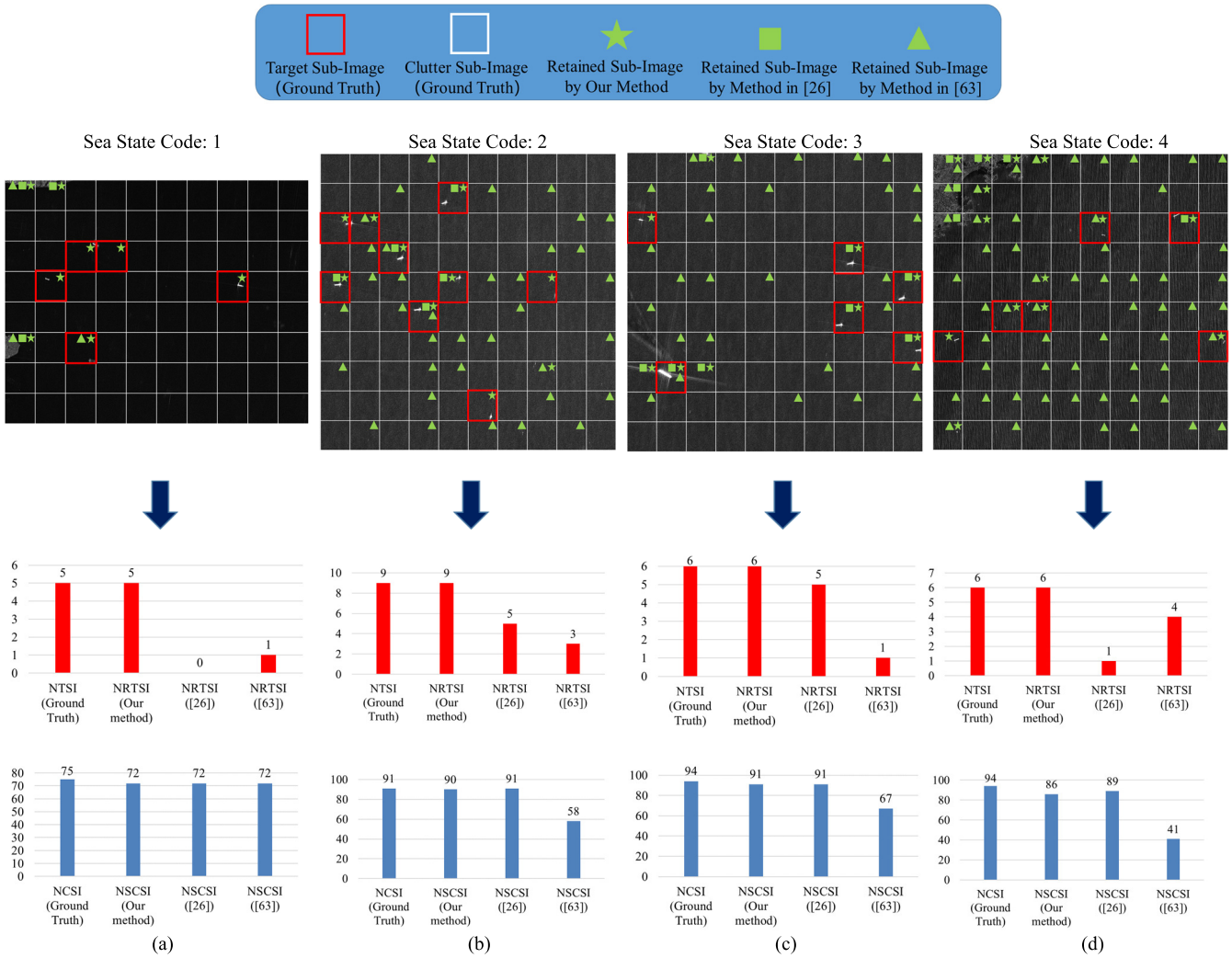


Fig. 13. Prescreening results of our density-based method (see Section III-A), the global CFAR method in [26], and the MESR method in [63] with (a) sea state code 1, (b) sea state code 2, (c) sea state code 3, and (d) sea state code 4. There are 2400 × 3000 pixels in (a) and 3000 × 3000 pixels in (b)–(d). In the first row, we show retained subimages by our method and those in [26] and [63]. In the second row, we compare numbers of retained target subimages by our method and [26], where "NTSI" and "NRTSI" denote the real number of target subimages and the number of retained target subimages, respectively. In the third row, we count numbers of screened clutter subimages by our method and those in [26] and [63], where "NCSI" and "NSCSI" denote the real number of clutter subimages and the number of screened clutter subimages, respectively.

stronger clutter level and lower quality of marine SAR images. In Fig. 13, we investigate the robustness of the proposed density-based prescreening operation in Section III-A versus the sea state code. The competitors are the prescreening methods in [26] and [63] that are based on global CFAR and MSER strategies, respectively. From Fig. 13, we can see that the proposed prescreening operation is more robust than [26] and [63] versus the sea state code. Our method retains all the target subimages with different sea state codes, while the methods in [26] and [63] discard some target information. This is because these methods in [26] and [63] do not exploit the sparse characteristic of ship targets in spatiality. In Fig. 14, we also show three retained subimages by our method from Fig. 13(a), (c), and (d), respectively, where there are relatively dark ship targets. Our density-based prescreening method successfully retains subimages with weak targets since they are still sparse (with low densities) in subimages. Note that we can use the sliding window strategy to estimate statistical distributions of the sea clutter in the retained subimages provided in Fig. 14.

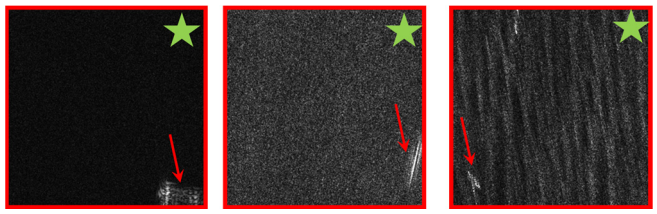


Fig. 14. Three subimages with relatively dark targets retained by our prescreening method from Fig. 13(a), (c), and (d), respectively. Red arrows denote dark targets.

In Table V, we compare the BRs, ASAs, and running times of MST [17], PILS [21], E-SLIC [24], SLIC [7], and the proposed DSLIC method using SAR images in Fig. 13. From Table V, we can see that the proposed DSLIC method

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TABLE V

COMPARISONS OF BRS, ASAS, AND RUNNING TIMES (SECONDS) USING SAR IMAGES IN FIG. 13, WHERE THE SUPERPIXEL SIZE IS $S = 18$

		MST [17]	PILS [21]	E-SLIC [24]	SLIC [7]	Proposed DSLIC
Sea state code 1	BR	0.6021	0.6316	0.6632	0.7811	0.8842
	ASA	0.1041	0.5903	0.6508	0.6710	0.7288
	Running time	496.7	4366	2166	2230	39.92
Sea state code 2	BR	0.3212	0.9350	0.8671	0.9178	0.9522
	ASA	0.0017	0.8868	0.8272	0.9153	0.9257
	Running time	787.6	6351	3931	3636	50.95
Sea state code 3	BR	0.1990	0.8146	0.6420	0.8059	0.8068
	ASA	0.0009	0.9050	0.8882	0.9138	0.9857
	Running time	746.6	5658	3288	3273	46.19
Sea state code 4	BR	0.7323	0.8640	0.9178	0.9249	0.9646
	ASA	0.1664	0.5813	0.8734	0.7404	0.8503
	Running time	835.7	5777	3309	3350	75.27

maintains its superiority in terms of segmentation performance and efficiencies with different sea state codes.

D. Case Study: Robustness of Superpixel-Based Segmentation Versus Ship Sizes

In this section, we investigate the robustness of our proposed method with different ship sizes. In Fig. 15(a), we plot the histogram of sizes of 1935 ship targets in 1723 SAR images [33]. Six examples in this dataset are shown in Fig. 15(b). As also reported in [33], ship targets in this dataset are multiscale. Fig. 15(c) illustrates the BR and ASA values of different methods with different superpixel sizes, where the proposed DSLIC method shows better or comparable segmentation performance in comparison with competitors in [7], [17], [21], and [24] based on multiscale ship targets. MST [17] provides deteriorated performance in this complex dataset. Note that the dense and uniform initialization of superpixel centers is used in DSLIC for Fig. 15(c) since the size of SAR image chips in [33] is small (i.e., 256 × 256), and it is not necessary to discard subimages with pure sea clutter therein. Fig. 15(c) demonstrates the effectiveness of the density proximity in (7).

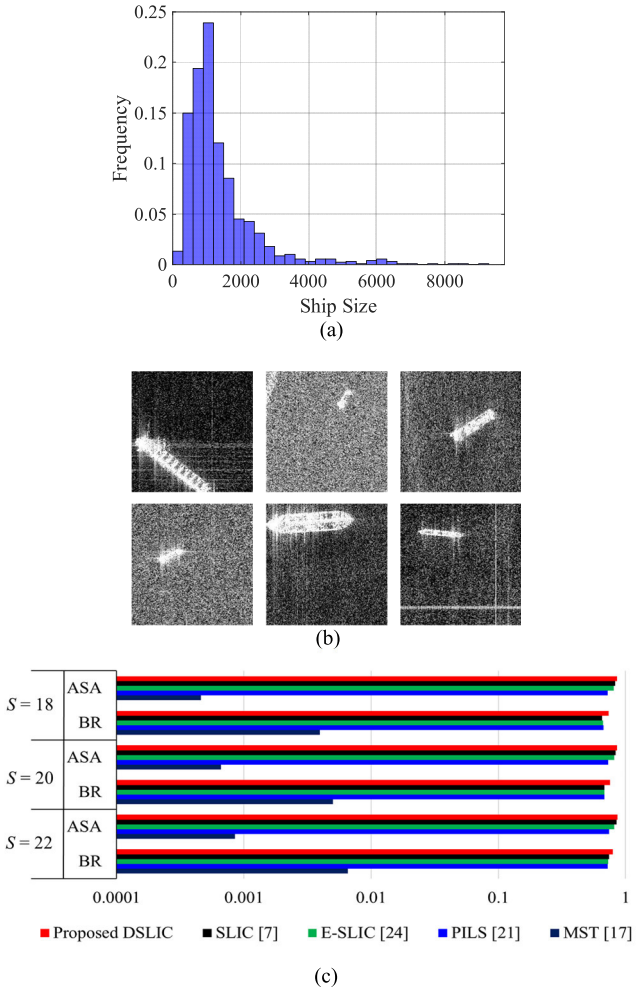


Fig. 15. (a) Histogram of the number of pixels occupied by 1935 ship targets in 1723 SAR images [33]. (b) Six examples in the SAR image dataset [33]. (c) BR and ASA values of MST [17], PILS [21], E-SLIC [24], SLIC [7], and the proposed DSLIC method with different superpixel sizes.

E. Case Study: Marine SAR Image With Large Size

1) Visualization of Superpixels: In Fig. 16, we compare the aforementioned superpixel generation algorithms by using a marine SAR image with a large size (7800 × 11 400 pixels). The resolution of the Sentinel-1 SAR image used in Fig. 16 is lower than that of Gaofen-3 SAR images in Figs. 6 and 10. The sea state code of Fig. 16(a) has not been provided in [32]. Fig. 16(a) shows six subimages (300 × 300 pixels) in the large image presented in Fig. 16(b), containing different numbers of ship targets. It can be observed from Figs. 16(c)–(f) that PILS [21], E-SLIC [24], and the proposed DSLIC method better capture the boundary pixels of ship targets than SLIC [7]. More regular and compact superpixels are obtained by DSLIC. Especially, the subimages preserved by the proposed density-driven classifier cover all the ship targets in both crowded and uncrowded areas. The preserved ratio in Fig. 15(f) is $\varepsilon = N^{(\text{perser})}/N^{(\text{subimg})} = 180/988 \approx 0.18 \ll 1$. The preserved subimages in Fig. 16(f) only include 4.8% (47/988) "false alarm subimages" occupied by the pure clutter. The results in Figs. 6, 10, and 16 also show the robustness of our method on the resolutions of SAR images since ship targets show lower densities than the sea clutter with different image resolutions. Note that the performance of MST [17] is not reported in Fig. 16 because the graph for Fig. 16(b) requires establishing a matrix with $(7800 \times 11\,400)^2 \approx 7.9 \times 10^{15}$ elements that exceed the limits of MATLAB (i.e., 2.8×10^{14} elements in a matrix).

2) Quantitative Metrics of Segmentation Performance: In Table VI, we compare the BR and ASA values using the SAR image in Fig. 16(b). We can see that the BR values of the proposed DSLIC method are comparable to those of other competitive approaches. DSLIC ranks first or second in the ASA metric and provides a significantly larger ASA value than SLIC [7]. This means that the proposed sparsity proximity

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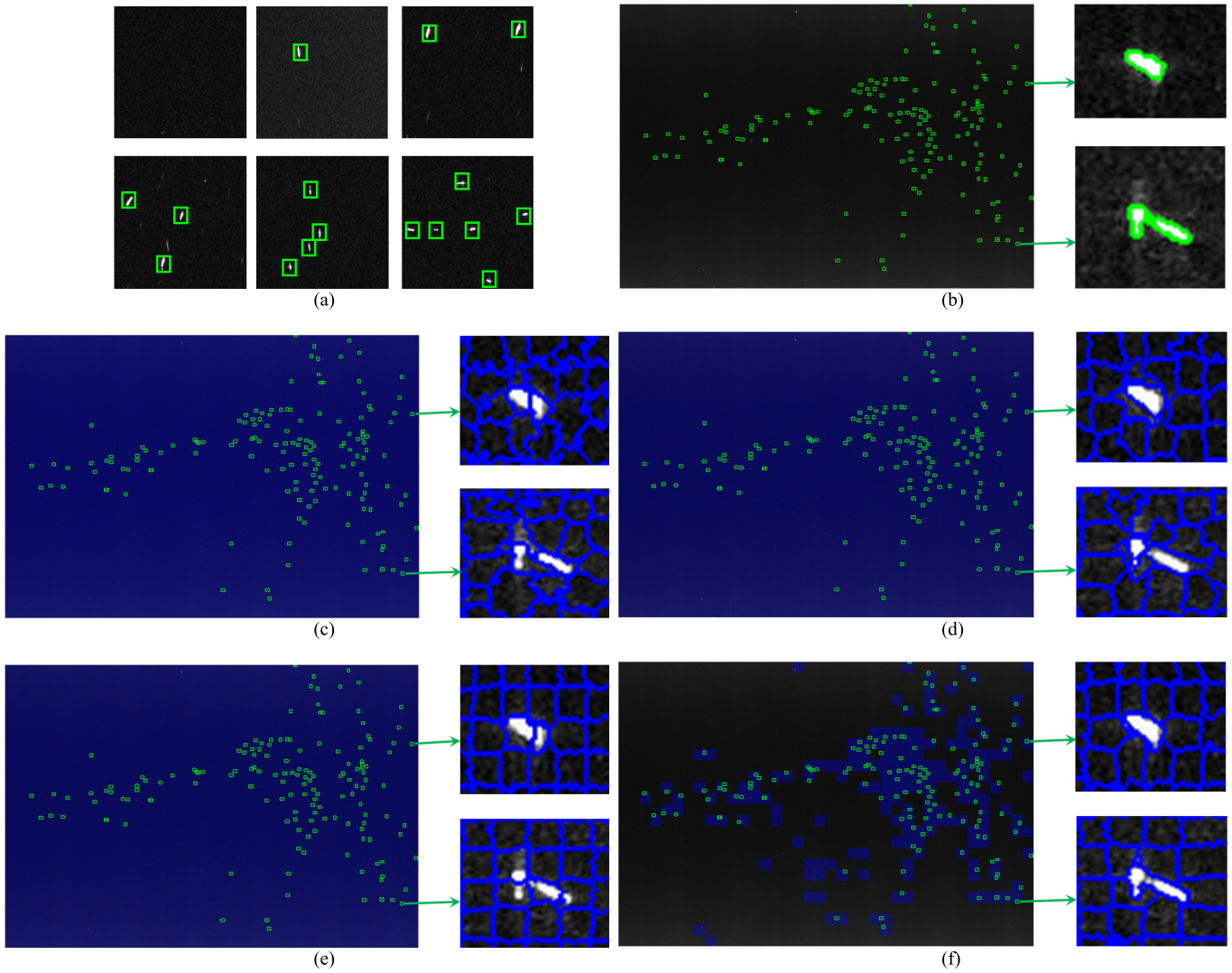


Fig. 16. (a) Six image chips in (b) Sentinel-1 SAR image with large size (7800 × 11 400 pixels). Superpixel-based segmentation results of (c) PILS [21], (d) E-SLIC [24], (e) SLIC [7], and (f) proposed DSLIC, where $S = 12$ and $T_{\max} = 3$. Blue lines are the boundaries of superpixels and green rectangles are used to locate ship targets. Green lines in (b) represent the human-labeled boundaries of ship targets.

TABLE VI

COMPARISON OF THE BR AND ASA VALUES USING THE SAR IMAGE IN FIG. 16(b)

Methods	BR		ASA	
	$S = 12$	$S = 14$	$S = 12$	$S = 14$
PILS [21]	0.9862	0.8943	0.7356	0.7004
E-SLIC [24]	0.9863	0.8870	0.8158	0.5065
SLIC [7]	0.9663	0.8094	0.6491	0.6673
Proposed DSLIC	0.9687	0.8903	0.7665	0.7361

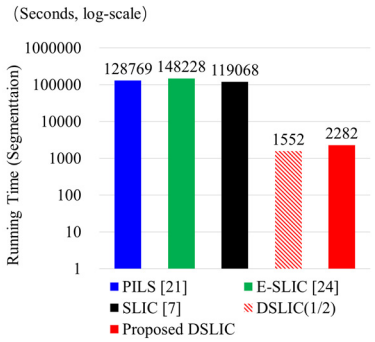


Fig. 17. Comparison of the running times of different superpixel-based segmentation methods, where the SAR image in Fig. 16(b) is used and the superpixel size is $S = 12$.

concept in DSLIC is helpful to enhance segmentation performance when processing SAR images.

3) Specific Segmentation Time: In Fig. 17, we summarize the running times of different segmentation methods performing on the image in Fig. 16(b). Similar to the cases reported in Figs. 9 and 12, the proposed DSLIC (1/2) and DSLIC require significantly smaller running times than other competitors. It is also worth mentioning that the proposed initialization stage in DSLIC can also be applied to accelerate other state-of-the-art SLIC-based methods, such as PILS [21] and E-SLIC [24].

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TABLE VII

AUC VALUES AND RUNNING TIMES OF THE SLCM DETECTOR [40] USING SUPERPIXELS GENERATED BY METHODS IN [7], [17], [21], AND [24] AND THE PROPOSED DSLIC METHOD. THE SAR IMAGE IN FIG. 16(b) IS USED, AND THE SUPERPIXEL SIZE IS $S = 12$

Methods	AUC	Running Time (Detection)
PILS [21]	0.7356	1.4×10^5
E-SLIC [24]	0.9906	1.2×10^5
SLIC [7]	0.9675	1.6×10^5
Proposed DSLIC	0.9829	31

4) Target Detection Performance and Complexity: In Table VII, we show the AUC values and running times of SLCM [40] using the large SAR image in Fig. 16(b). From Table VII, we can see that the AUC of DSLIC is higher than that of SLIC [7]. This is because the new sparsity proximity in DSLIC enhances the compactness of superpixels in marine SAR images. Besides, the AUC of DSLIC is close to that of the state-of-the-art E-SLIC [24] but only requires 0.03% of its running time. This demonstrates the superiority of the proposed DSLIC method in terms of computation efficiency when processing large marine SAR images.

VI. CONCLUSION AND DISCUSSIONS

In this article, a new algorithm called DSLIC has been proposed for the superpixel-based segmentation of marine SAR images. In our newly developed DSLIC, density features are exploited based on the sparsity of ship targets in terms of spatiality. In the initialization stage of DSLIC, a density-driven classifier is designed via the WSLMS strategy to prescreen each subregion of a large SAR image and remove most of the pure clutter areas. This alleviates the computational complexity and memory requirements of the subsequent local clustering stage and enforces a target-aware superpixel-based segmentation. In the local clustering stage of DSLIC, the concept of sparsity proximity of pixels (reflected by the density distances) is newly incorporated to improve the representation of pixel proximity and enhance segmentation performance. Theoretical and experimental results demonstrate that the proposed DSLIC method is faster, more efficient in terms of memory usage, and able to provide comparable or better segmentation performance compared with other existing state-of-the-art approaches. Specifically: 1) the segmentation time of DSLIC is 1%–10% of those of state-of-the-art approaches and 2) the superpixel-based detection time using superpixels generated by DSLIC is 0.02%–0.3% of those using those generated by state-of-the-art approaches.

In this article, we mainly consider superpixel-based segmentation as a fast prescreening stage of SAR ship monitoring. We also expect an end-to-end and well pretrained deep learning model to be able to realize prescreening, detection, and classification of ship targets. However, current end-to-end and well pretrained deep learning models suffer from the lack of complete data labeled with high quality. End-to-end deep learning models for SAR ship monitoring require a significant number of large SAR images with both ground-truth locations and categories of ship targets therein. To the best of our knowledge, the open dataset [32] is the closest available one to meet the aforementioned requirements. However, the open dataset [32] still has a few limitations to achieve end-to-end deep learning models for SAR ship monitoring. For example, as shown in [32, Fig. 5], the available categories of ship targets are significantly unbalanced. It is difficult to train a good subnetwork for ship classification in end-to-end deep learning models with significantly unbalanced data. Therefore, existing research works often consider the preprocessing, detection, and classification stages according to the size and quality of available labeled data for each stage.

We also notice that Xu *et al.* [67] reported that the Chinese HISEA-1 satellite has achieved onboard SAR ship detection based on the combination of CFAR methods and pretrained deep learning methods. However, to the best of our knowledge, the accomplishment of onboard SAR ship classification has not been fully reported. It is our introspection that this is because open, well-balanced, and abundant labeled datasets for onboard SAR ship classification are currently not available.

The combination of the proposed DSLIC method and onboard deep-learning-based ship detection methods [67] together with the extraction of additional shape information [61] of ship targets using superpixels will be pursued in our future research.

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